

Exercise

SQL Fundamentals: SELECT & FILTERING

① Selecting & Sorting

① Retrieve all columns from the employees table.

Id	First-name	Last-name	department	Salary	Hire-date	City
1	Alice	Green	IT	70000	2010-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-08	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

SELECT *

FROM employees

② Find all unique departments

SELECT distinct(department)

FROM employees

employees

department

IT

HR

Finance

Marketing

③ Retrieve first and last names ordered by salary descending.

SELECT first-name

last-name

FROM employees

ORDER BY salary DESC

employees

first-name	last-name
Eva	Nardoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

④ Retrieve the top 3 highest-paid employees.

SELECT id,

first-name,

last-name,

Salary

FROM employees

WHERE salary = highest

ORDER BY salary DESC

LIMIT 3

employees

id	first-name	last-name	Salary
5	Eva	Nardoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Q2 Filtering with WHERE

⑤ Find employees in the IT department

```
SELECT id,  
       first-name,  
       last-name,  
       department
```

FROM employees

WHERE department = 'IT'

employees

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Nardoo	IT

⑥ Find employees in finance with Salary > 60000

```
SELECT id,  
       first-name  
       last-name  
       department  
       salary
```

FROM employees

WHERE salary > 60000 AND department = 'finance'

employees

id	first-name	last-name	department	salary
3	Cathy	Zulu	finance	65000

⑦ Find employees in HR or Marketing

```
SELECT id,  
       first-name,  
       last-name,  
       department
```

FROM employees

WHERE department = 'HR' OR 'Marketing'

employees

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Motoena	Marketing

⑧ Find employees not in IT

```
SELECT id,  
       first-name  
       last-name  
       department
```

FROM employees

WHERE employees NOT = 'IT'

employees

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	finance
4	David	Motoena	marketing

⑨ Find employees in IT, HR OR finance using IN

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department IN('IT', 'HR', 'finance')
```

employees

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	finance
5	Eva	Naidoo	IT

⑩ Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id,  
       first-name,  
       last-name,  
       department,  
       salary,  
       city  
FROM employees  
WHERE department = 'IT' with salary > 65000 AND city = 'Johannesburg'
```

employees

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg